

■ 2.5.7 Formatting *Mathematica* Programs

When you write programs using procedures and loops, it is very important that you format them in a way that makes their structure clear. It is usually a good idea to put each element of a procedure on a separate line. You should indent the lines to indicate what loop or procedure they are in.

```

ToCycles[perm_?PermutationQ] :=
  Block[{a, t, n, l, i, len},
    len = Length[perm];
    a = {} ;
    t = Table[True, {len}];
    For[i=1, i<=len, i++,
      If[t[[i]],
        For[n = perm[[i]]; l = {}, t[[n]], n = perm[[n]],
          t[[n]] = False; AppendTo[l, n]
        ];
        AppendTo[a, l]
      ]
    ] ;
  Return[a]
]

```

A *Mathematica* program, illustrating formatting conventions.

Even if you just have a sequence of nested functions, it is often a good idea to put successive nesting levels on different lines, with progressively larger indents. If the sequence of nested functions is short enough, you can put them all on one line, but add spaces to emphasize the nesting.

```

FromCycles0[list_, c_] :=
  Block[{c1},
    c1 = RotateRight[c,1];
    Table[ list[ c1[[i]] ] = c[[i]], {i, 1, Length[c]} ]
]

```

A *Mathematica* definition, illustrating the use of spaces in nested functions.